Spectroscopic Comparison between NGC 2392 (Eskimo Nebula) & its Host Star

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ABSTRACT

This study explores the spectroscopic differences between the dying host star of the NGC 2392 (Eskimo Nebula) and its surrounding planetary nebula using near-infrared spectroscopy.

Observations were conducted at the Perkins Telescope Observatory using the Mimir instrument (a near infrared wide-field imager, spectrometer, and polarimeter). Raw spectral images were obtained by employing a 2-pixel wide slit in HK-Spec observing mode, which utilizes a 1.40 μm long-pass filter in conjunction with the JHK grism. This setup allowed for detailed spectroscopy in the H and K bands of both the host star and the planetary nebula.

Results indicate the detection of hydrogen transition levels in both the host star and the nebula. However, the planetary nebula also exhibits significant amounts of helium (He I), distinguishing its composition from that of the host star. These findings provide key insights into the evolution of dying stars and the distinct characteristics of the gas ejected from them during the late stages of stellar life.

The study contributes to a better understanding of the physical processes involved in the formation of planetary nebulae and their chemical differentiation from their progenitor stars.

From Star to Nebula

When a star dies that usually means that it is running low on energy. The fusion that keeps this star active is dying out. Thus the star is entering its final stages. When a star enters these final stages it begins to shed its outer layers. These outer layers expand and glow with very hot gas; creating a *planetary nebula*. Although nebulae form from their dying host stars they can be distinguishable in spectra. Depending on how distinguishable nebulae and hosts are we can make theoretical predictions about the star's final stages and expansion of the nebula.

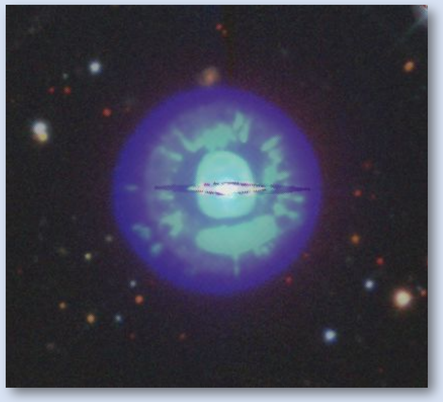


Fig 1. NGC 2392 (Eskimo Nebula).

Taken using Aladin Sky Atlas. Base image from ALLWISE Survey. With layers of DESI-Legacy, PanSTARRS, and 2MASS sky surveys.

METHODS

We used Mimir, a near-infrared wild-field imager, spectrometer, and polarimeter, at Perkins Telescope Observator, owned by Boston University. Prof. Dan Clemens and I obtained raw spectral images by inserting a 2-pixel (1.2 arcsec) wide by 5 arcmin tall slit at the center of the Mimir field of view. After we inserted the decker to reject all light not coming through the slit and selecting the HK-Spec observing mode. This mode commands the filter wheels to insert a 1.40 um long-pass filter, which rejects all light with wavelength shorter than 1.40 um, and the JHK grism, the combination of a prism and a transmission grating. The JHK grism is the optical element that disperses the light along the X-axis of the Raw Spectral Images.

CONCLUSIONS

Although NGC 2392(Eskimo Nebula) is not very distinguishable from its dying host star we can still make theoretical predictions for the both celestial objects. The spectra I gathered specifies that the host star is producing enough ultraviolet radiation to emit hydrogen and helium into the nebula. Since the star spectra is deficit of helium I can infer that the star is likely close to the end of its planetary nebula phase, where it will soon shrink into a white dwarf. As it begins to cool and lose mass it may never achieve the high energy levels seen in its planetary nebula surrounding it. The strong bracket and helium lines in the nebula implies a well developed planetary nebula in an advanced stage of evolution. Of course it will continue to expand and leave a faint shell around its white dwarf.

RESULTS

Fig 2. Spectra of Eskimo Central Star

This spectroscopy graph of Eskimo Nebula Host Star represents the massive amounts of hydrogen. The graph shows hydrogen in a bracket series where electrons are transitioning from higher to lower energy levels as the wavelength goes up. The brackett series also suggest the star is cooler than others but still hot enough to expel hydrogen ions at high energy levels. The gaps and noise in spectra likely result from stellar winds, pulsations and/or circumstellar gas; further research required.

Fig 3. Spectra of Eskimo Nebula

Similar to its host star the nebula spectra presents high amounts of hydrogen; also in transitioning in bracket series. The Nebula does present singular ionized helium lines which is common in planetary nebula. The hydrogen and helium indicates that the nebula is high energy environment which is consistent with its host star. This environment likely pairs with ongoing ionization as the nebula expands.

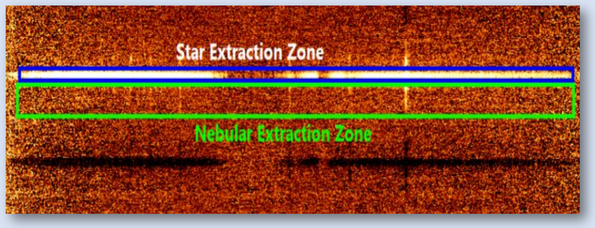


Fig 4. NGC 2392 (Eskimo Nebula Spectra) zones

Two horizontal bands to represent the the spectra zones of both Fig. 1 & Fig. 2. The blue zone indicates the host star spectral zone. It's more concentrated and narrow to focus on capturing the star's light. The green zone indicate the nebula. Since nebulas are more extended and diffused; this zone is more broad. By distinguishing these zones I was able to study the star and nebula independently; providing more insight into the stellar characteristics and ionized gases of the nebulae.

REFERENCES

This research has made use of "Aladin sky atlas" developed at CDS, Strasbourg Observatory, France
2000A&AS..143...33B (Aladin Desktop), 2014ASPC..485..277B (Aladin Lite v2), and 2022ASPC..532....7B (Aladin Lite v3).

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ACKNOWLEDGEMENTS

I would like to give thanks to Dr. Athina Meli (NC A&T) and Dr. Dan Clemens (Boston University) for their mentorship on this project and to the NSF Partnerships in Astronomy & Astrophysics Research and Education (PAARE) grant.

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